

Git Hands-On Lab – Branching and Merging

1. Creation and Switching to New Branch

```
MINGW64:/c/Users/JONITH J V/GitDemo

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~
$ cd ~/GitDemo

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    welcome.txt~

no changes added to commit (use "git add" and/or "git commit -a")

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (main)
$ git branch GitNewBranch

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (main)
$ git checkout GitNewBranch
D       welcome.txt~
Switched to branch 'GitNewBranch'
```

Step 2: Create a New File

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (GitNewBranch)
$ echo "This file was created in GitNewBranch." > branch.txt

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (GitNewBranch)
$ git add branch.txt
warning: in the working copy of 'branch.txt', LF will be replaced by CRLF the next time Git touches it

JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (GitNewBranch)
$ git commit -m "Added branch.txt in GitNewBranch"
[GitNewBranch bf5af3e] Added branch.txt in GitNewBranch
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt

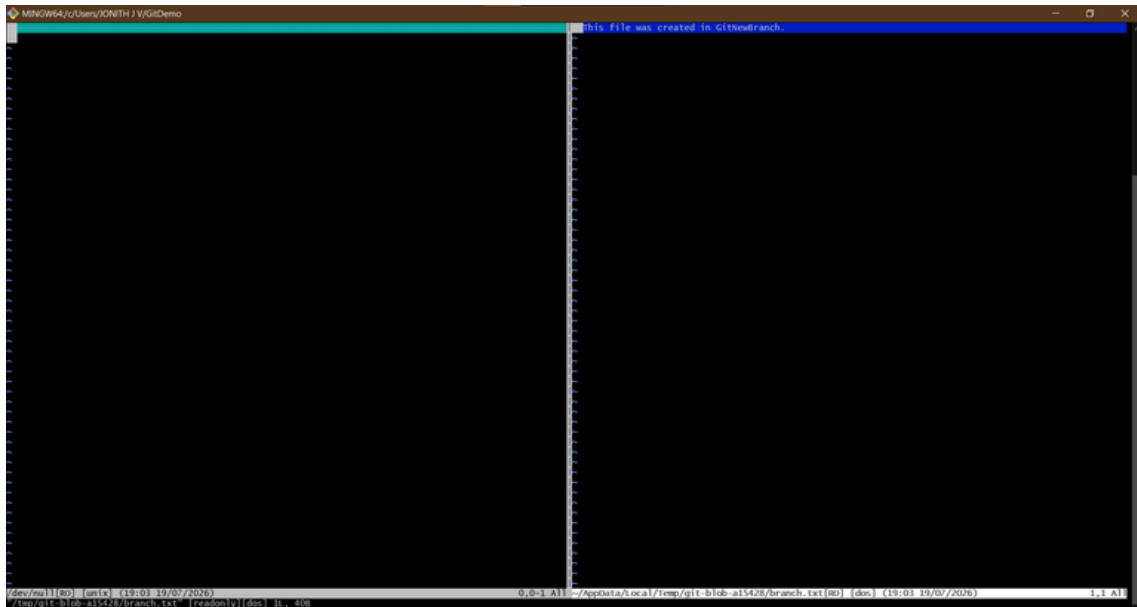
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    welcome.txt~

no changes added to commit (use "git add" and/or "git commit -a")
```

Step 3 : Merging

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (GitNewBranch)
$ git switch main
D       welcome.txt~
Switched to branch 'main'
Your branch is ahead of 'origin/master' by 3 commits.
  (use "git push" to publish your local commits)
```

Step 4: Compare the Branches



Step 5: Merged the Branch

```
$ git merge GitNewBranch
Updating ec3f33e..bd5625c
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt
```

Step 6: Viewed the Commit History

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/GitDemo (main)
$ git branch
git status
git log --oneline --graph --decorate
* main
On branch main
Your branch is ahead of 'origin/master' by 4 commits.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add/rm <file>.." to update what will be committed)
  (use "git restore <file>.." to discard changes in working directory)
        deleted:    welcome.txt~

no changes added to commit (use "git add" and/or "git commit -a")
* bf5af3e (HEAD -> main, origin/main) Added branch.txt in GitNewBranch
* ec6d867 Successfully configured .gitignore for log files and folders
* 87c25e5 Merge remote changes
|\
| * 9dfb736 Initial commit
|/
* 97f3d5a (origin/master) Initial Commit
```

Step 7: Deletion of Branch

```
JONITH J V@LAPTOP-93Q0MAID MINGW64 ~/Desktop/coding/Git/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was bd5625c).
```